SUPPLEMENTARY APPENDIX

This appendix has been provided by the authors to give readers additional information about their work.

Supplement to: Bardia A, Rugo HS, Tolaney SM, et al. Final Results From the Randomized Phase III ASCENT Clinical Trial in Metastatic Triple-Negative Breast Cancer and Association of Outcomes by HER2 and Trop-2 Expression

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Supplemental Methods

Inclusion Criteria

1. Female or male patients, aged > 18 years, able to understand and give written informed consent.

2. Histologically or cytologically confirmed triple-negative breast cancer (TNBC) per ASCO/CAP criteria, based on the most recent analyzed biopsy or other pathology specimen. Triple negative is defined as < 1% expression for estrogen receptor (ER) and progesterone receptor (PR) and negative

for human epidermal growth factor receptor 2 (HER2) by in situ hybridization.

3. Metastatic disease documented by CT or MRI imaging.

4. Measurable disease by CT or MRI as per RECIST 1.1. Bone-only disease is not permitted.

5. Brain MRI must be done for patients with brain metastasis and patient must have had stable* CNS disease for at least 4 weeks.*A maximum of 15% (N = 74) of patients with brain metastases will be included in this trial.

Prior local treatment by radiation, surgery, or stereotactic surgery

- Imaging – stable or decreasing size after such local treatment
- Clinically stable signs and symptoms
- ≥ 2 weeks from discontinuation of anti-seizure medication.

Corticosteroid (if needed) - dose should be stable or decreasing for at least 2 weeks before randomization. Steroid dose should be 20 mg or less of prednisone/prednisolone daily, or equivalent of a different steroid.

6. At least 2 weeks beyond high-dose systemic corticosteroids (however, low-dose corticosteroids ≤ 20 mg prednisone or equivalent daily are permitted provided the dose is stable for 4 weeks).

7. Refractory to or relapsed after at least two prior standard of care chemotherapy regimens for unresectable, locally advanced, or metastatic breast cancer. (These regimens will qualify regardless of triple-negative status at the time they were given.)

There is no cap on the number of prior chemotherapies for locally advanced or metastatic disease and earlier adjuvant or neoadjuvant therapy for more limited disease will qualify as one of the required prior regimens if the development of unresectable, locally advanced, or metastatic disease occurred within a 12-month period of time after completion of chemotherapy).

a. For patients with a documented germ-line *BRCA1/BRCA2* mutation who received an approved PARP inhibitor, the PARP inhibitor can be used to meet the criteria for one of two prior standard of care chemotherapies.

b. All patients must have been previously treated with a taxane regardless of disease stage (adjuvant, neoadjuvant, or advanced) when it was given. Patients who have contra-indications or are intolerant to taxanes are eligible provided that they received at least one cycle of a taxane and showed contra-indications or intolerance during or at the end of that cycle.

8. Eligible for one of the chemotherapy options listed as treatment of physician's choice (TPC) (eribulin, capecitabine, gemcitabine, or vinorelbine) as per investigator assessment.

9. Eastern Cooperative Oncology Group performance score of 0 or 1.

10. Adequate hematology without transfusion support (hemoglobin > 9 g/dL, ANC > 1,500

per mm^3, platelets > 100,000 per mm^3). Blood transfusion or growth factor support is not allowed within 14 days prior to screening labs.

11. Adequate renal and hepatic function (creatinine clearance of > 60 mL/min, may be

calculated using Cockcroft-Gault equation; bilirubin ≤ 1.5 IULN, AST and ALT ≤ 2.5 × IULN or ≤ 5 × IULN if known liver metastases and serum albumin ≥ 3 g/dL).

12. Recovered from all toxicities to grade 1 or less by NCI CTCAE v4.03 (except alopecia

or peripheral neuropathy that may be grade 2 or less) at the time of randomization.

Patients with grade 2 neuropathy are eligible but may not receive vinorelbine as TPC.

13. Patients must have completed all prior cancer treatments at least 2 weeks* prior to randomization including chemotherapy (includes also endocrine treatment), radiotherapy and major surgery.

14. Prior investigational agents are permitted, provided completion according to the timeframes above.

15. Patients must have a life expectancy of 3 months or greater, in the opinion of the investigator.

*Prior antibody treatment for cancer must have been completed at least 3 weeks prior to randomization.

Exclusion Criteria

1. Women who are pregnant or lactating.

2. Women of childbearing potential or fertile men unwilling to use highly effective* contraception during study and up to 3 months after treatment discontinuation in women of child-bearing potential and 6 months in males post last study drug.

3. Patients with Gilbert's disease.

4. Patients with non-melanoma skin cancer or carcinoma in situ of the cervix are eligible, while patients with other prior malignancies must have had at least a 3-year disease-free interval.

5. Patients known to be HIV positive

6. Patients with hepatitis B positive or hepatitis C positive infection.**

7. Known history of unstable angina, MI, or CHF present within 6 months of randomization or clinically significant cardiac arrhythmia (other than stable atrial fibrillation) requiring anti-arrhythmia therapy.

8. Known history of clinically significant active COPD, or other moderate-to-severe chronic respiratory illness present within 6 months of randomization.

9. Prior history of clinically significant bleeding, intestinal obstruction, or GI perforation within 6 months of randomization.

10. Infection requiring antibiotic use within 1 week of randomization.

11. Patients with active chronic inflammatory bowel disease (ulcerative colitis, Crohn's disease) and patients with a history of bowel obstruction.

12. Patients who have received a live vaccine within 30 days of randomization.

13. Patients who previously received irinotecan.

14. Rapid deterioration during screening prior to randomization, eg, significant change in ECOG performance status, 20% decrease in serum albumin levels, unstable pain symptoms requiring modifications in analgesic management.

15. Other concurrent medical or psychiatric conditions that, in the Investigator's opinion, may be likely to confound study interpretation or prevent completion of study procedures and follow-up examinations.

*Highly effective is defined as Combined (estrogen and progestogen containing) contraception associated with inhibition of ovulation: oral, injectable, implantable, intrauterine device (IUD), intrauterine hormone-releasing system (IUS), bilateral tubal occlusion, vasectomized partner, sexual

abstinence). Abstinence refers to "True abstinence," which means it is in line with the preferred and usual lifestyle of the patient. Periodic abstinence (eg, calendar, ovulation, symptothermal, post-ovulation methods), declaration of abstinence for the duration of exposure to study treatment, and withdrawal are not acceptable methods of contraception.

**In patients with a history of HBV, hepatitis B core antibody (HBcAb) testing is required and

if positive, then HB DNA testing will be performed and if positive the patient will be excluded.

Patients

For detailed information on inclusion/exclusion criteria, please see above. Briefly, eligible patients had relapsed/refractory metastatic TNBC (according to the standard ASCO/CAP criteria; HER2 IHC0, 1, or 2/ISH-negative; ER/PR < 1%)[1-3] after ≥ 2 prior standard chemotherapy regimens (≥ 1 in the metastatic setting; no upper limit) for unresectable, locally advanced, or metastatic disease. Patients with known HER2-positive disease were ineligible for ASCENT; HER2-negative status (HER2 IHC0 or IHC1+; if IHC2+, then ISH-negative) was based on local assessment of the most recent biopsy or pathology report. Trop-2 expression was not required to determine patient eligibility. Patients with known brain metastases (capped at 15%).

Trial Design and Treatment

The ASCENT study design has been previously described.[4] Patients were randomly assigned 1:1 to receive either sacituzumab govitecan (SG; 10 mg/kg intravenously on days 1 and 8, every 21 days) or TPC (capecitabine, eribulin, vinorelbine, or gemcitabine) until disease progression, unacceptable toxicity, study withdrawal, or death, whichever occurred first. No crossover to the SG group was allowed upon progression with chemotherapy, with no restrictions on subsequent therapies after study discontinuation. Stratification factors at randomization included number of prior chemotherapy regimens for advanced disease (2 to 3 *v* > 3), presence of known brain metastases at baseline (yes *v* no), and geography (North America *v* rest of world).

End points

The primary endpoint was progression-free survival (PFS) per RECIST 1.1 by blinded independent central review (BICR) in patients who were BMNeg. Secondary end points included PFS in the intention-to-treat (ITT) population by BICR, objective response rate (ORR) by BICR, overall survival (OS), and safety.[5] Exploratory end points included evaluating Trop-2 expression in tumor tissue and post-hoc analyses included subgroup analysis of outcomes by HER2 status.

Subgroup Analyses

Trop-2 expression was determined using primary or metastatic archival biopsies or surgical specimens using a validated IHC assay as previously reported.[6] Trop-2 expression was categorized based on a histochemical score (H-score), a numerical value represented by a weighted summation of percent staining which accounts for both the staining intensity and the percentage of cells at that intensity. Categories were established by the following staining intensity scores: 0 indicated no staining (I0); 1+ indicated weak staining (I1); 2+ indicated moderate staining (I2); and 3+ indicated intense staining (I3). Membrane H-scores were then defined as the proportion of Trop-2 expressing cells weighted by their intensities (H-score = $I1 + 2 \times I2 + 3 \times I3$) and divided into equally sized quartiles. Trop-2 expression was also assessed using percent membrane cell staining of I2+I3, which was the proportion of cells with intensities I2 and I3. They were then divided into four equally sized quartiles. Samples were assessed centrally.

For the current post hoc HER2 subgroup analysis, local IHC and ISH results for the ITT population of ASCENT were analyzed retrospectively to associate SG efficacy with HER2 status as HER2 IHC0 or HER2-low, defined as HER2 IHC1+, or IHC2+ and ISH-negative.

Thus, among the patients in the ITT, only patients with a known Trop-2 expression or HER2 IHC status were included in these subgroup analyses.

Statistical Analyses

For the current exploratory analyses, nominal *P* values are reported.[4,7] Kaplan–Meier estimates were used to analyze PFS and OS between each treatment group, with medians and corresponding 95% CIs determined according to the Brookmeyer and Crowley method with log-log transformation. The magnitude of the PFS and OS benefit was measured by hazard ratios (HRs) and their 95% CIs estimated from stratified Cox proportional-hazards models. Stratified log-rank tests were used to

obtain *P* values of treatment effect. Response rates between the treatment groups were compared using the stratified Cochran-Mantel-Haenszel method.

Efficacy was evaluated in all randomized patients (ITT population). Safety was assessed in patients who received one or more doses of study drug. Key statistical assumptions were as previously described.[4]

For the post-hoc subgroup efficacy analyses, only patients with known Trop-2 expression or HER2 IHC status in the ITT population were included. For each subgroup, analyses for PFS and OS were conducted using a similar statistical approach for the ITT population. HRs of SG relative to TPC were estimated using an unstratified Cox proportional hazards model. The ORR and 95% CI were calculated based on the Clopper-Pearson method. The odds ratios of SG relative to TPC for ORR were estimated the using Mantel-Haenszel method.

The final database lock date was February 25, 2021, which included further efficacy and safety data in the ITT population, and by HER2 status and Trop-2 expression collected from the 17 patients who remained in the trial after the final data cutoff of March 11, 2020.

FIG S1. ASCENT Study Design (NCT02574455).

Adapted from *N Engl J Med*. Bardia A, Hurvitz SA, Tolaney SM, et al. Sacituzumab govitecan in metastatic triple-negative breast cancer. Vol. 384, pp 1529-1541. Copyright ©2022 Massachusetts Medical Society.[4] Reused with permission from Massachusetts Medical Society.

[a] PFS measured by an independent, centralized, and blinded group of radiology experts who assessed tumor response using RECIST 1.1 criteria in patients without brain metastasis.

[b] The ITT population includes all randomized patients (with and without brain metastases). Baseline brain MRI only required for patients with known brain metastasis.

ASCO/CAP, American Society of Clinical Oncology/College of American Pathologists; DOR, duration of response; ITT, intention-to-treat; IV, intravenous; MRI, magnetic resonance imaging; ORR, objective response rate; OS, overall survival; PFS, progression-free survival; QoL, quality of life; R, randomization; RECIST 1.1, Response Evaluation Criteria in Solid Tumors, version 1.1; TNBC, triple-negative breast cancer; TTR, time to response.

National Institutes of Health. https://clinicaltrials.gov/ct2/show/NCT02574455.

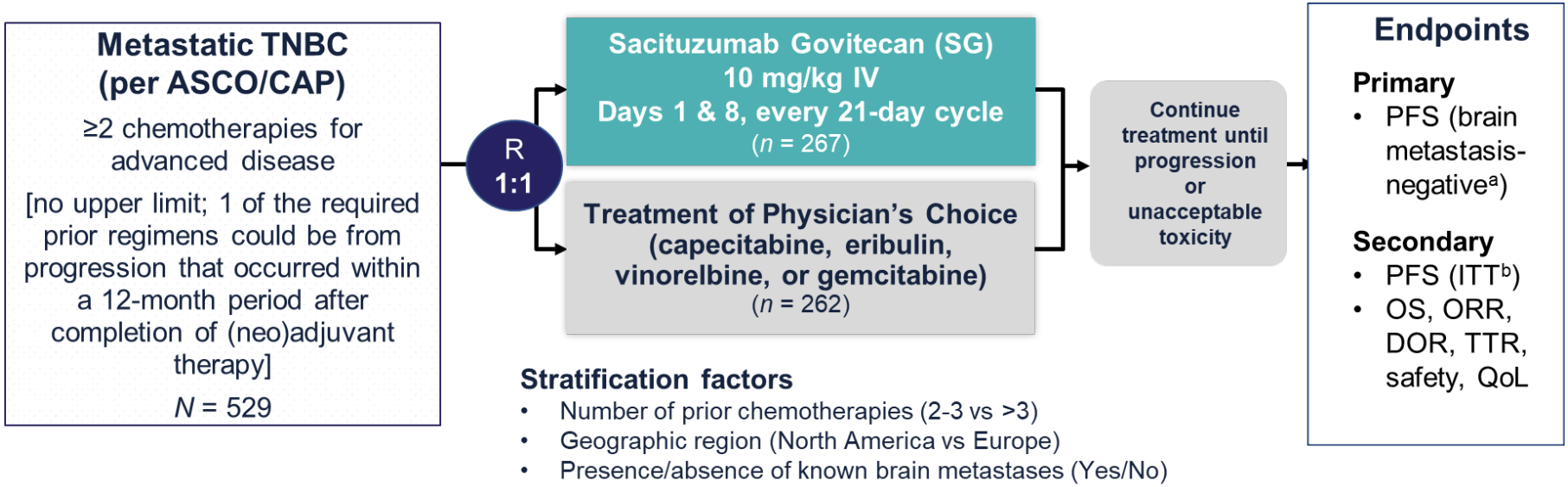
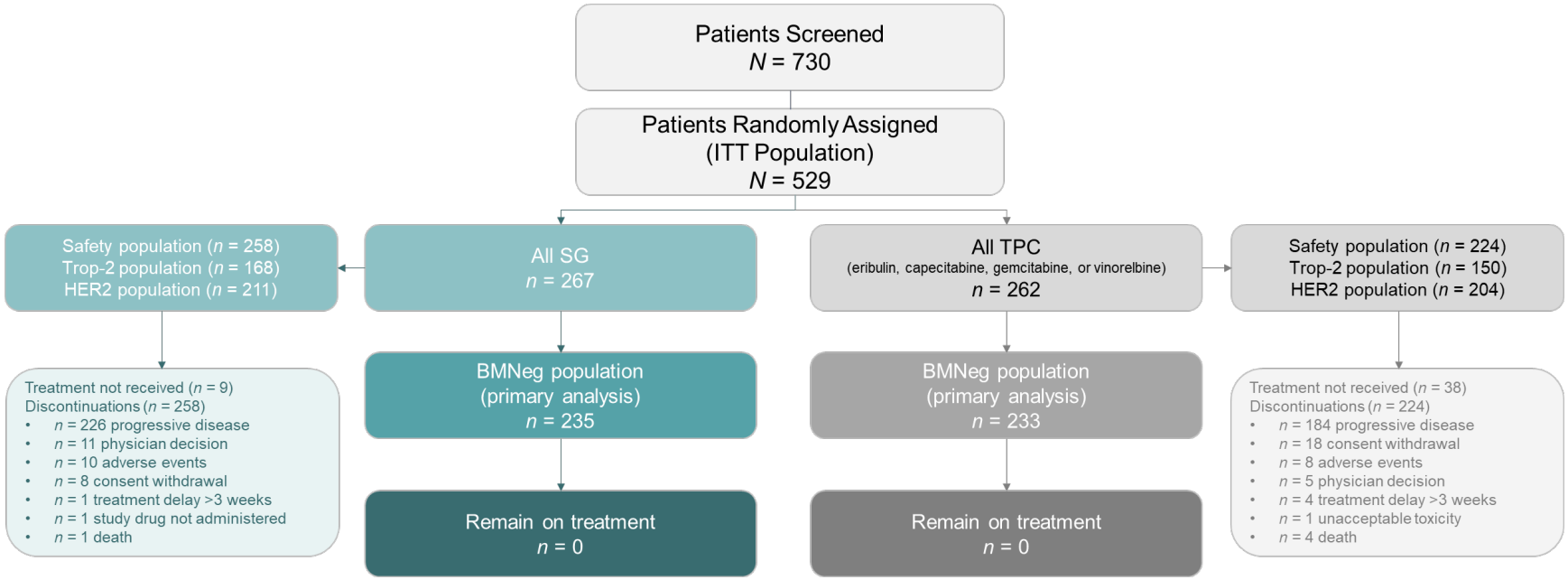


FIG S2. CONSORT Diagram - Patient Disposition.

One death in the sacituzumab govitecan arm was due to respiratory failure and was considered unlikely related to treatment. Patients in the TPC arm were randomly assigned to eribulin (n = 139), vinorelbine (n = 52), gemcitabine (n = 38), or capecitabine (n = 33). BMNeg, brain metastases-negative; HER2, human epidermal growth factor 2; ITT, intention-to-treat; SG, sacituzumab govitecan; TPC, treatment of physician's choice Trop-2, tumor-associated calcium signal transducer 2.



Median follow-up was calculated from time of randomization to death or last known alive date

TABLE S1. Response Rates (ITT population)

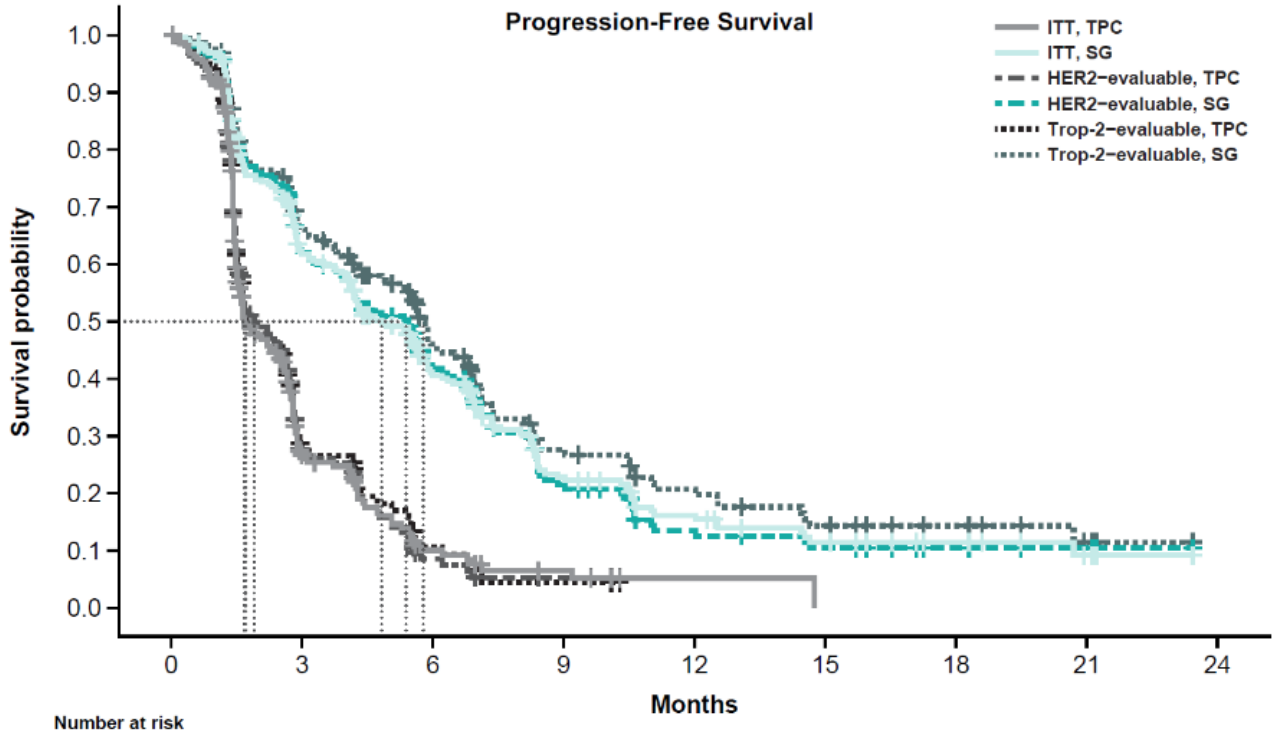
	ITT Population	
	SG (n=267)	TPC (n=262)
ORR, n (%)	83 (31)	11 (4)
OR (95% CI), nominal *P*-Value[a]	11.0 (5.7-21.4), *P*< .0001	
Best overall response, n (%)[b]		
CR	10 (4)	2 (1)
PR	73 (27)	9 (3)
CBR, n (%)[c]	108 (40)	21 (8)
OR (95% CI), nominal *P*-Value[a]	8.1 (4.8-13.5), *P*< .0001	
Median DOR, mo (95% CI)[d]	6.3 (5.5-7.9)	3.6 (2.8-NE)
Median TTR, mo (range)[d]	1.5 (0.7-10.6)	1.5 (1.3-4.2)

[a] The *P* value is based on the Cochran-Mantel-Haenszel test. [b] Objective response is defined as the best confirmed overall response of either CR or PR. The best overall response is derived based on independent review assessed tumor response at each tumor assessment according to RECIST 1.1. Responses of CR and PR are confirmed no less than 4 weeks later. SD requires a minimum duration of 6 weeks to be classified as SD. [c] CBR is defined as the percentage of patients with a confirmed best overall response of CR or PR, and SD with a duration of at least 6 months. [d] Only patients achieving CR or PR are included in the analysis of DOR and TTR. Median DOR is from Kaplan-Meier estimate. CI for median is computed using the Brookmeyer-Crowley method. Abbreviations: CR, complete response; ITT, intention-to-treat; NE, not evaluable; ORR, objective response rate; PD, progressive disease; PR, partial response; SD, stable disease; SG, sacituzumab govitecan; TPC, treatment of physician's choice; Trop-2, tumor-associated calcium signal transducer 2.

FIG S3. Progression-Free Survival and Overall Survival (ITT population and Subgroups).

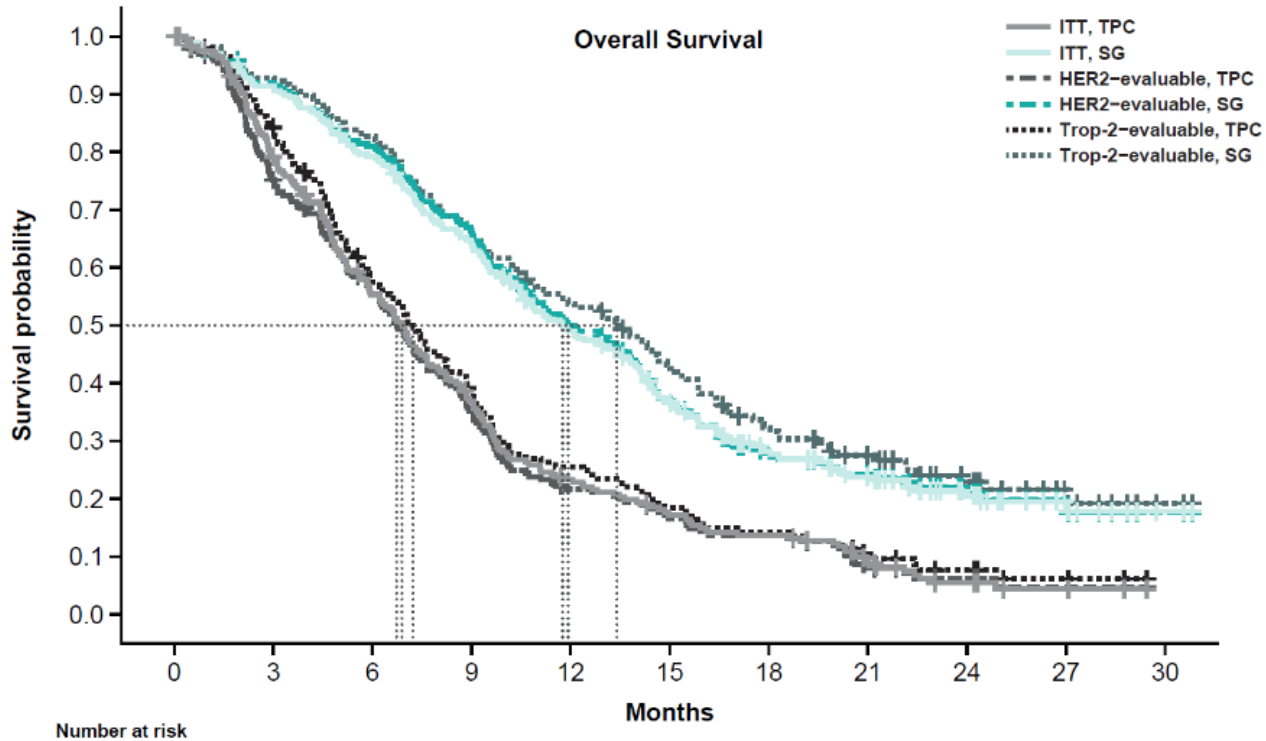
Median progression-free survival and overall survival from Kaplan–Meier estimates in ITT population and biomarker evaluable subgroups. PFS is defined as the time from the date of randomization to the date of the first radiological disease progression or death due to any cause, whichever comes first. OS is defined as the time from date of randomization to the date of death from any cause. Patients without documentation of death are censored on the date they were last known to be alive. CI for median PFS and OS was computed using the Brookmeyer-Crowley method. Stratified log-rank test and stratified Cox regression was adjusted for the following stratification factors: number of prior chemotherapies and region. HER2, human epidermal growth factor 2; ITT, intention-to-treat; OS; overall survival; PFS, progression-free survival; SG, sacituzumab govitecan; TPC, treatment of physician's choice; Trop-2, trophoblast cell surface antigen 2.

A



Number at risk									
ITT, TPC	262	40	12	5	1	0	0	0	0
ITT, SG	267	145	82	38	23	14	8	3	0
HER2−evaluable, TPC	204	33	8	3	0	0	0	0	0
HER2−evaluable, SG	211	115	66	28	14	10	5	3	0
Trop-2−evaluable, TPC	150	25	8	2	0	0	0	0	0
Trop-2−evaluable, SG	168	100	59	30	20	13	8	3	0

B



Number at risk											
ITT, TPC	262	192	132	87	54	39	31	16	7	3	0
ITT, SG	267	242	209	169	125	92	62	42	25	11	2
HER2−evaluable, TPC	204	141	102	65	39	29	24	11	6	2	0
HER2−evaluable, SG	211	193	169	138	101	73	49	36	21	9	2
Trop-2−evaluable, TPC	150	121	81	55	36	26	20	11	7	3	0
Trop-2−evaluable, SG	168	155	138	110	88	69	49	34	21	9	2

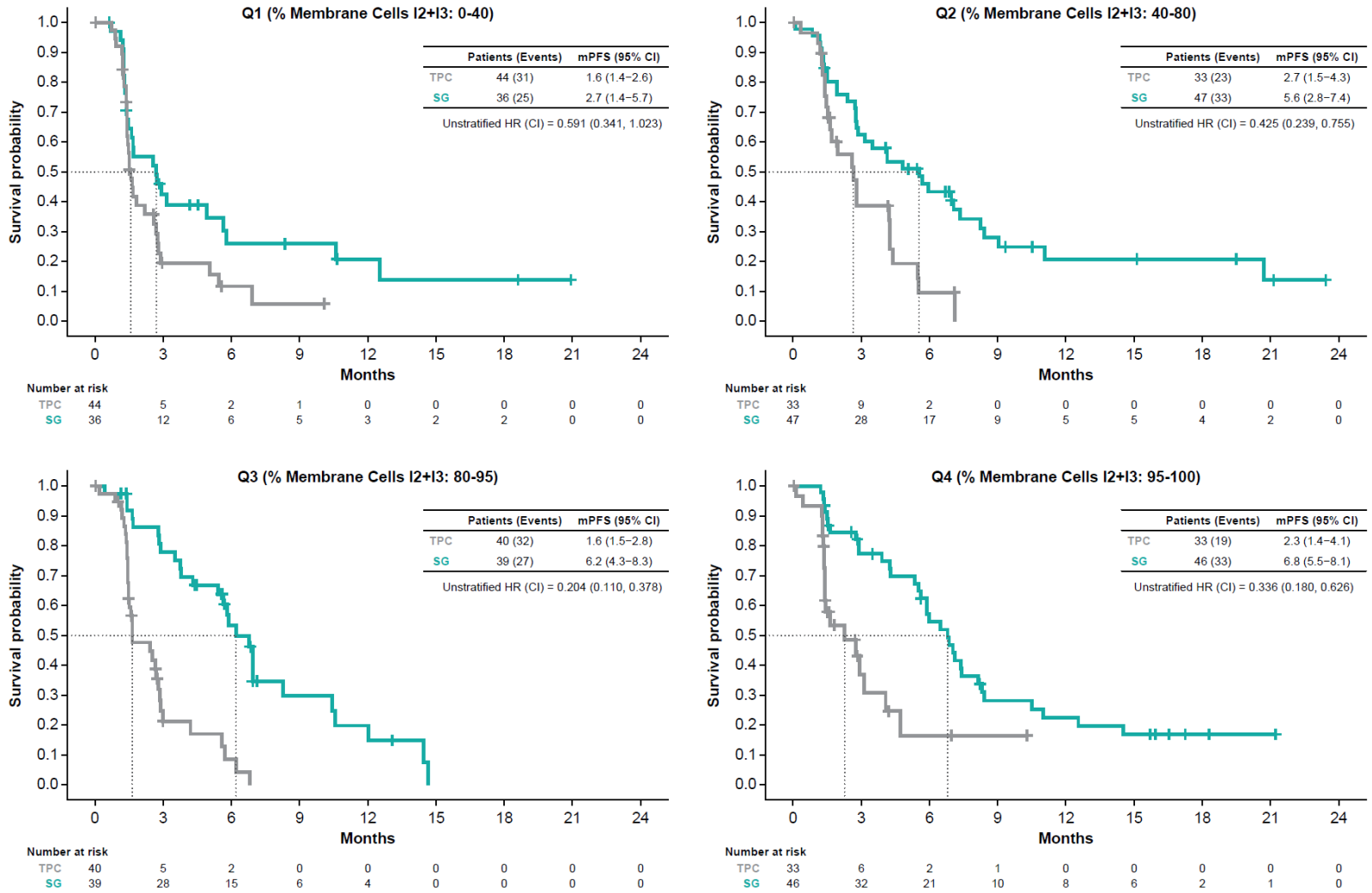
TABLE S2. Efficacy Outcomes in HER2 and Trop-2 Subgroups (ITT population)

	HER2 Status[a]				Trop-2 Expression (H-Score)								Trop-2 Expression (% membrane cells)							
	HER2 IHC0		HER2-Low[b]		H-score: 0-130		H-score: 130-220		H-score: 220-275		H-score: 275-300		% Membrane Cells 0-40		% Membrane Cells 40-80		% Membrane Cells 80-95		% Membrane Cells 95-100	
	SG (n=149)	TPC (n=144)	SG (n=62)	TPC (n=60)	SG (n=35)	TPC (n=45)	SG (n=47)	TPC (n=33)	SG (n=39)	TPC (n=40)	SG (n=47)	TPC (n=32)	SG (n=36)	TPC (n=44)	SG (n=47)	TPC (n=33)	SG (n=39)	TPC (n=40)	SG (n=46)	TPC (n=33)
Median PFS[c] (95% CI)	4.3 (3.0-5.8)	1.6 (1.5-2.4)	6.2 (3.8-7.1)	2.9 (1.6-4.2)	2.7 (1.4-5.7)	1.5 (1.4-2.2)	4.8 (2.9-7.1)	2.8 (1.7-4.3)	6.8 (4.3-8.3)	1.6 (1.4-2.7)	6.9 (5.6-8.1)	2.8 (1.4-3.1)	2.7 (1.4-5.7)	1.6 (1.4-2.6)	5.6 (2.8-7.4)	2.7 (1.5-4.3)	6.2 (4.3-8.3)	1.6 (1.5-2.8)	6.8 (5.5-8.1)	2.3 (1.4-4.1)
Hazard ratio for disease progression or death (95% CI)	0.38 (0.28-0.50)		0.45 (0.27-0.73)		0.58 (0.34-1.00)		0.52 (0.29-0.92)		0.20 (0.11-0.36)		0.31 (0.17-0.59)		0.59 (0.34-1.02)		0.43 (0.24-0.76)		0.20 (0.11-0.38)		0.34 (0.18-0.63)	
Median OS[d] (95% CI)	11.7 (9.9-14.0)	5.9 (4.8-7.3)	13.4 (9.6-15.2)	8.7 (6.7-9.7)	8.7 (6.9-12.9)	7.0 (4.9-9.6)	13.4 (7.8-16.5)	8.8 (4.8-10.2)	15.2 (11.8-17.5)	6.5 (4.1-8.2)	14.5 (10.6-18.3)	7.1 (4.9-9.8)	8.8 (6.9-12.9)	7.0 (4.9-9.6)	14.5 (7.4-17.8)	7.4 (3.9-10.1)	15.2 (11.8-17.5)	7.7 (5.9-9.1)	14.2 (10.5-17.0)	5.9 (4.7-9.8)
Hazard ratio for disease progression or death (95% CI)	0.50 (0.39-0.65)		0.52 (0.34-0.78)		0.74 (0.46-1.20)		0.69 (0.41-1.17)		0.34 (0.21-0.57)		0.36 (0.22-0.59)		0.70 (0.43-1.13)		0.65 (0.38-1.11)		0.36 (0.22-0.60)		0.35 (0.21-0.58)	

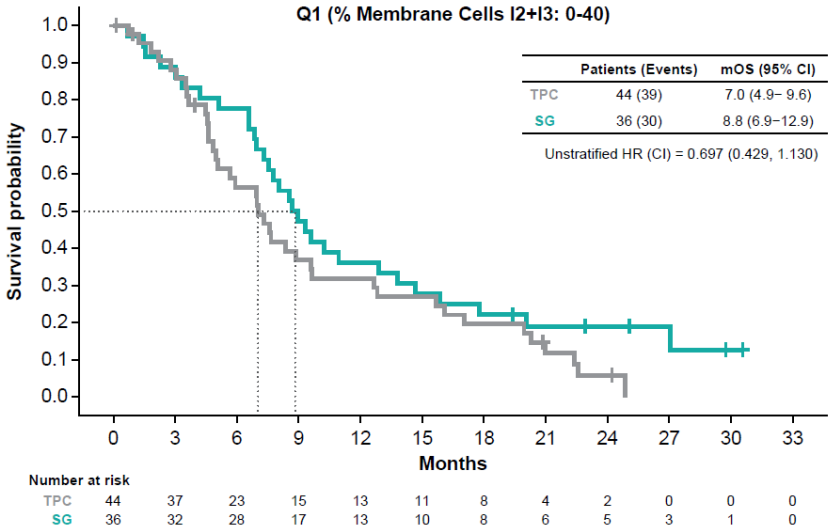
[a] HER2 CI for median PFS and median OS was computed using the Brookmeyer-Crowley method. Stratified log-rank test and stratified Cox regression was adjusted for the following stratification factors: number of prior chemotherapies and region.[b] Defined as IHC1+, IHC 2+/ISH-negative. [c] PFS is defined as the time from the date of randomization to the date of the first radiological disease progression or death due to any cause, whichever comes first. [d] OS is defined as the time from date of randomization to the date of death from any cause. Patients without documentation of death are censored on the date they were last known to be alive.
Abbreviations: ITT, intention-to-treat; HER2, human epidermal growth factor 2; H-score, histochemical score; IHC, immunohistochemistry; SG, sacituzumab govitecan; TPC, treatment of physician's choice.

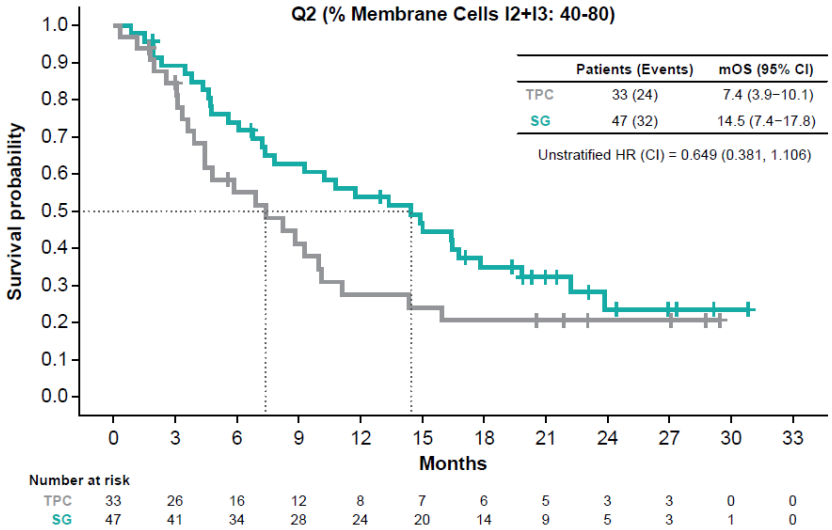
FIG S4. Progression-Free Survival and Overall Survival in Patients by Trop-2 Expression, % Membrane Staining (ITT population).

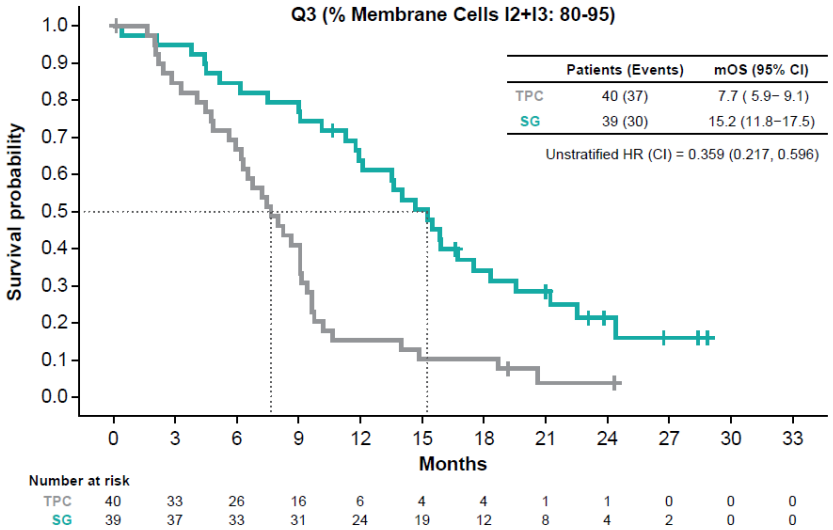
A. Progression-Free Survival



B. Overall Survival







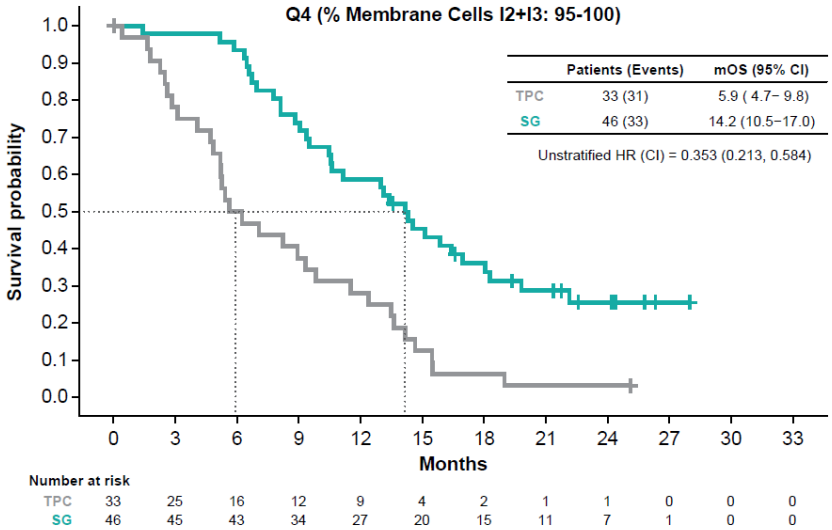


TABLE S3. Response Rates in HER2 and Trop-2 Subgroups (ITT population)

	HER2 Population				Trop-2 Population (H-Score)								Trop-2 Population (% membrane cells)							
	HER2 IHC0		HER2-Low[a]		H-score: 0-130		H-score: 130-220		H-score: 220-275		H-score: 275-300		% Membrane Cells 0-40		% Membrane Cells 40-80		% Membrane Cells 80-95		% Membrane Cells 95-100	
	SG (n=149)	TPC (n=144)	SG (n=62)	TPC (n=60)	SG (n=35)	TPC (n=45)	SG (n=47)	TPC (n=33)	SG (n=39)	TPC (n=40)	SG (n=47)	TPC (n=32)	SG (n=36)	TPC (n=44)	SG (n=47)	TPC (n=33)	SG (n=39)	TPC (n=40)	SG (n=46)	TPC (n=33)
ORR, n (%)	46 (31)	5 (4)	20 (32)	5 (8)	8 (23)	2 (4)	13 (28)	5 (15)	16 (41)	0	21 (45)	0	8 (22)	2 (5)	16 (34)	5 (15)	13 (33)	0	21 (46)	0
OR (95% CI)	12.4 (4.8-32.3)		5.2 (1.8-15.1)		6.4 (1.3-32.3)		2.1 (0.7-6.7)		NE		NE		6.0 (1.2-30.4)		2.9 (0.9-8.9)		NE		NE	
Best overall response, n (%)[b]																				
CR	3 (2)	0	3 (5)	1 (2)	3 (9)	0	2 (4)	2 (6)	3 (8)	0	2 (4)	0	3 (8)	0	2 (4)	2 (6)	2 (5)	0	3 (7)	0
PR	43 (29)	5 (3)	17 (27)	4 (7)	5 (14)	2 (4)	11 (23)	3 (9)	13 (33)	0	19 (40)	0	5 (14)	2 (5)	14 (30)	3 (9)	11 (28)	0	18 (39)	0
CBR, n (%)[c]	54 (36)	9 (6)	30 (48)	7 (12)	8 (23)	4 (9)	20 (43)	6 (18)	22 (56)	2 (5)	26 (55)	2 (6)	8 (22)	4 (9)	23 (49)	6 (18)	19 (49)	2 (5)	26 (57)	2 (6)
OR (95% CI)	8.5 (4.0-18.1)		7.1 (2.8-18.0)		3 .0 (0.8-11.1)		3.3 (1.2-9.6)		24.6 (5.2-116.6)		18.6 (4.0-86.8)		2.9 (0.8-10.4)		4.3 (1.5-12.4)		18.0 (3.8-85.4)		20.2 (4.3-94.4)	
Median DOR, mo (95% CI)[d]	6.9 (5.4-9.0)	2.9 (2.8-NE)	5.6 (4.3-NE)	3.6 (2.9-NE)	11.3 (3.5-NE)	NE	6.9 (2.8-NE)	3.6 (2.9-NE)	9.0 (5.5-12.3)	NE	5.6 (4.2-9.8)	NE	11.3 (3.5-NE)	NE	7.1 (3.2-NE)	3.6 (2.9-NE)	5.6 (4.2-9.2)	NE	5.6 (4.4-NE)	NE
Median TTR, mo (95% CI)[d]	1.6 (1.4-2.8)	1.4 (1.3-NE)	1.5 (1.4-3.3)	1.4 (1.3-NE)	1.5 (1.2-1.5)	1.4 (1.3-NE)	2.8 (1.4-3.3)	1.4 (1.3-NE)	1.5 (1.4-4.2)	NE	1.6 (1.5-4.2)	NE	1.4 (1.2-1.5)	1.4 (1.3-NE)	2.8 (1.4-4.0)	1.4 (1.3-NE)	1.4 (1.4.-3.0)	NE	1.5 (1.5-4.2)	NE

[a] Defined as IHC1+, IHC 2+/ISH-negative. [b] Objective response is defined as the best confirmed overall response of either CR or PR. The best overall response is derived based on independent review assessed tumor response at each tumor assessment according to RECIST 1.1. Responses of CR and PR are confirmed no less than 4 weeks later. SD requires a minimum duration of 6 weeks to be classified as SD. [c] CBR is defined as the percentage of patients with a confirmed best overall response of CR or PR, and SD with a duration of at least 6 months. [d] Only patients achieving CR or PR are included in the analysis of DOR and TTR. Median DOR is from Kaplan-Meier estimate. CI for median is computed using the Brookmeyer-Crowley method.

Abbreviations: CBR, clinical benefit rate; CR, complete response; DOR, duration of response; HER2, human epidermal growth factor 2; H-score, histochemical score; IHC, immunohistochemistry; ITT, intention-to-treat; NE, not evaluable; OR, odds ratio; ORR, objective response rate; PR, partial response; RECIST 1.1, Response Evaluation Criteria in Solid Tumors, version 1.1; SD, stable disease; SG, sacituzumab govitecan; TPC, treatment of physician's choice; Trop-2, trophoblast cell surface antigen 2; TTR, time to response.

TABLE S4. Objective Response Rate in Patients with Low Trop-2 Expression

	Low Trop-2 Expression							
	H-score = 0		H-score <50		% Membrane Cells = 0		% Membrane Cells <10	
	SG (n=7)	TPC (n=4)	SG (n=20)	TPC (n=21)	SG (n=9)	TPC (n=8)	SG (n=14)	TPC (n=17)
ORR, n (%)	1 (14)	0	3 (15)	2 (10)	2 (22)	0	3 (21)	0
OR (95% CI)	NE		1.7 (0.2-11.3)		NE		NE	
Best overall response, n (%)[b]								
CR	0	0	1 (5)	0	0	0	1 (7)	0
PR	1 (14)	0	2 (10)	2 (10)	2 (22)	0	2 (14)	0
CBR, n (%)[c]	1 (14)	0	3 (15)	4 (19)	2 (22)	1 (13)	3 (21)	2 (12)
OR (95% CI)	NE		0.8 (0.1, 3.9)		2.0 (0.1-27.4)		2.0 (0.3-14.4)	
Median DOR, mo (95% CI)[d]	NE	NE	9.3 (9.3-NE)	NE	9.3 (NE)	NE	9.3 (9.3-NE)	NE
Median TTR, mo (range)[d]	1.4 (NE)	NE	1.4 (1.4-NE)	1.4 (1.3-NE)	1.4 (1.4-NE)	NE	1.4 (1.4-NE)	NE

[a] The *P* value is based on the Cochran-Mantel-Haenszel test. [b] Objective response is defined as the best confirmed overall response of either CR or PR. The best overall response is derived based on independent review assessed tumor response at each tumor assessment according to RECIST 1.1. Responses of CR and PR are confirmed no less than 4 weeks later. SD requires a minimum duration of 6 weeks to be classified as SD. [c] CBR is defined as the percentage of patients with a confirmed best overall response of CR or PR, and SD with a duration of at least 6 months. [d] Only patients achieving CR or PR are included in the analysis of DOR and TTR. Median DOR is from Kaplan-Meier estimate. CI for median is computed using the Brookmeyer-Crowley method.

Abbreviations: CR, complete response; ITT, intention-to-treat; HER2, human epidermal growth factor 2; IHC, immunohistochemistry; NE, not evaluable; ORR, objective response rate; PD, progressive disease; PR, partial response; SD, stable disease; SG, sacituzumab govitecan; TPC, treatment of physician's choice.

FIG S5. Trop-2 Expression by HER2 Status

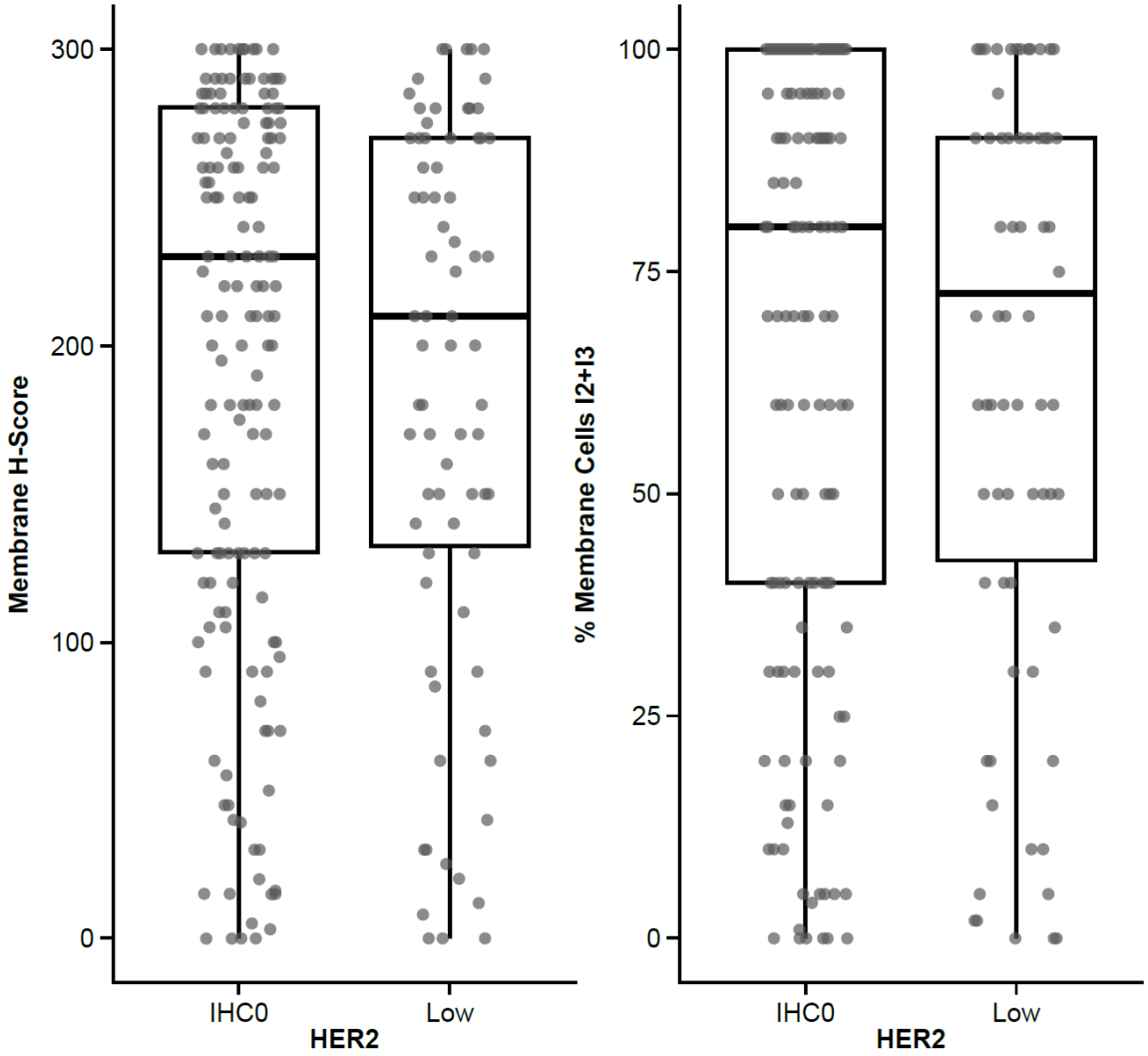


FIG S6. Progression-Free Survival and Overall Survival (ITT HER2-Evaluable Population).

Seventy-eight percent of the ITT population was HER2 evaluable by IHC. SG improved PFS and OS. Abbreviations: mOS, median overall survival; Q, quartile; SG, sacituzumab govitecan; TPC, treatment of physician's choice.

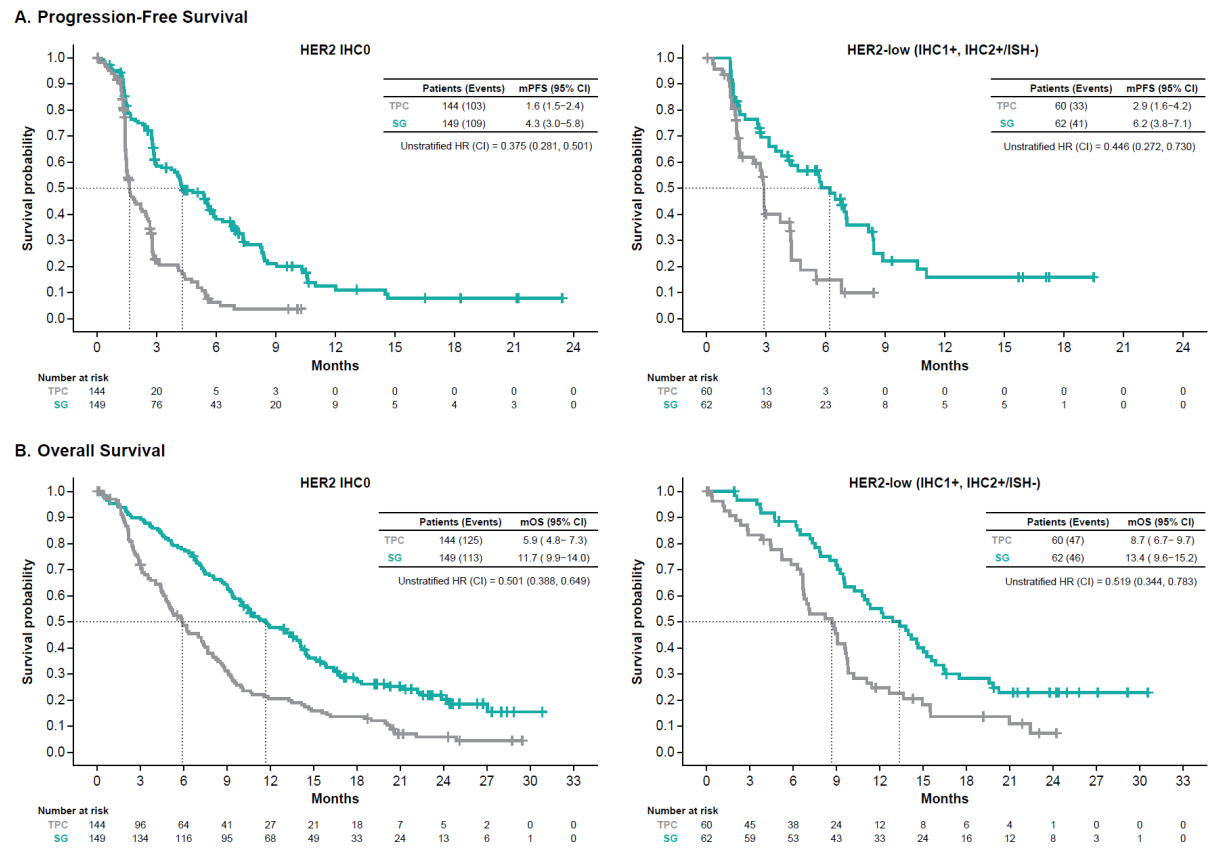


TABLE S5. Interaction of Trop-2 PFS and OS by treatment

Interaction analysis	P-values for interaction (Type III Wald chi-square test)	
	PFS (BICR)	OS
Treatment by Trop-2 H-Score Quartiles	0.244	0.054
Treatment by Trop2 % Membrane Cells I2+I3 Quartiles	0.726	0.190
Treatment by Continuous Trop-2 H-Score	0.366	0.039
Treatment by Continuous Trop-2 % Membrane Cells I2+I3	0.470	0.046

p-value for interaction was computed using type III Wald chi-square test

TABLE S6. Safety Summary

	Overall Safety Population		HER2 Population				Trop-2 Population (H-score)								Trop-2 Population (% membrane cells)							
			HER2 IHC0		HER2-low[a]		H-score: 0-130		H-score: 130-220		H-score: 220-275		H-score: 275-300		% Membrane Cells 0-40		% Membrane Cells 40-80		% Membrane Cells 80-95		% Membrane Cells 95-100	
	SG (n=258)	TPC (n=224)	SG (n=143)	TPC (n=119)	SG (n=60)	TPC (n=52)	SG (n=35)	TPC (n=44)	SG (n=47)	TPC (n=32)	SG (n=38)	TPC (n=37)	SG (n=47)	TPC (n=30)	SG (n=36)	TPC (n=42)	SG (n=47)	TPC (n=32)	SG (n=38)	TPC (n=39)	SG (n=46)	TPC (n=30)
Any TEAE,[b] n (%)	257 (100)	219 (98)	23 (16)	10 (8)	7 (12)	51 (98)	35 (100)	43 (98)	46 (98)	32 (100)	38 (100)	36 (97)	47 (100)	30 (100)	36 (100)	41 (99)	46 (98)	32 (100)	38 (100)	39 (100)	46 (100)	29 (97)
Grade ≥3	188 (73)	145 (65)	98 (66)	75 (52)	49 (82)	36 (69)	27 (77)	27 (61)	31 (66)	23 (72)	32 (84)	21 (57)	31 (66)	19 (63)	29 (81)	26 (62)	31 (66)	21 (66)	28 (74)	26 (67)	33 (72)	17 (57)
Leading to dose reduction	57 (22)	59 (26)	30 (21)	27 (23)	15 (25)	19 (37)	7 (20)	12 (27)	9 (19)	12 (38)	9 (24)	5 (14)	15 (32)	8 (27)	7 (19)	12 (29)	10 (21)	10 (31)	7 (18)	8 (21)	16 (35)	7 (23)
Leading to dose delay	162 (63)	87 (39)	88 (62)	43 (36)	39 (65)	20 (39)	24 (69)	17 (39)	27 (57)	10 (31)	24 (63)	12 (32)	32 (68)	13 (43)	26 (72)	16 (38)	27 (57)	10 (31)	20 (53)	15 (39)	34 (74)	11 (37)
Leading to treatment discontinuation	12 (5)	12 (5)	4 (3)	6 (5)	5 (8)	2 (4)	2 (6)	2 (5)	3 (6)	3 (9)	2 (5)	2 (5)	1 (2)	0	3 (8)	2 (5)	2 (4)	2 (6)	2 (5)	3 (8)	1 (2)	0
Leading to death	1 (<1)[c]	3 (1)	1 (<1)	0	0	3 (6)	0	1 (2)	0	0	0	1 (3)	0	0	0	1 (2)	0	0	0	1 (3)	0	0

[a] Defined as IHC1+, IHC 2+/ISH-negative. [b] The denominator for percentages is the number of patients in the Safety Population for each treatment group in each subgroup. TEAE is defined as an adverse event with start date on or after the date of first dose of study treatment and up to 30 days after date of last dose of study treatment. [c] This was considered unlikely to be related to SG treatment.

Abbreviations: HER2, human epidermal growth factor 2; H-score, histochemical score; SG, sacituzumab govitecan; TEAE, treatment-emergent adverse event; TPC, treatment of physician's choice; Trop-2, trophoblast cell surface antigen 2

TABLE S7. Treatment Emergent Adverse Events (Safety Population; All Grade, ≥ 10%; Grade ≥ 3, ≥ 5% of Patients)

TEAE[a]		SG (n=258)		TPC (n=224)	
		All grade (%)	Grade ≥ 3 (%)	All grade (%)	Grade ≥ 3 (%)
Hematologic	Neutropenia[b]	165 (64)	135 (52)	98 (44)	76 (34)
	Anemia[c]	103 (40)	24 (9)	62 (28)	13 (6)
	Leukopenia[d]	43 (17)	27 (11)	27 (12)	13 (6)
	Lymphopenia[e]	26 (10)	5 (2)	13 (6)	5 (2)
	Thrombocytopenia[f]	17 (7)	4 (2)	28 (13)	5 (2)
	Febrile Neutropenia	15 (6)	15 (6)	6 (3)	6 (3)
Hepatic	Aspartate aminotransferase increased	30 (12)	7 (3)	27 (12)	6 (3)
	Alanine aminotransferase increased	28 (11)	3 (1)	22 (10)	3 (1)

Gastrointestinal	Diarrhea	168 (65)	30 (12)	38 (17)	2 (1)
	Nausea	161 (62)	8 (3)	68 (30)	1 (< 1)
	Vomiting	87 (34)	5 (2)	36 (16)	3 (1)
	Constipation	96 (37)	1 (< 1)	52 (23)	0
	Abdominal Pain	54 (21)	7 (3)	18 (8)	3 (1)
	Stomatitis	27 (11)	2 (1)	14 (6)	0
Infections	Urinary Tract Infection	35 (14)	1 (<1)	18 (8)	1 (< 1)
	Upper Respiratory Tract Infection	32 (12)	0	7 (3)	0
Other	Fatigue	133 (52)	11 (4)	89 (40)	19 (9)
	Pyrexia	39 (15)	1 (< 1)	32 (14)	5 (2)
	Asthenia	40 (16)	4 (2)	29 (13)	3 (1)
	Edema Peripheral	25 (10)	0	23 (10)	2 (1)
	Decreased Appetite	71 (28)	4 (2)	46 (21)	2 (1)

Hypokalemia	46 (18)	7 (3)	29 (13)	1 (< 1)
Hypomagnesemia	32 (12)	0	13 (6)	0
Back Pain	45 (17)	3 (1)	31 (14)	4 (2)
Arthralgia	41 (16)	2 (1)	21 (9)	0
Headache	47 (18)	2 (1)	28 (13)	1 (< 1)
Dizziness	27 (11)	0	16 (7)	0
Insomnia	30 (12)	0	11 (5)	0
Cough	61 (24)	0	40 (18)	1 (< 1)
Dyspnea	45 (17)	10 (4)	47 (21)	12 (5)
Alopecia	121 (47)	0	36 (16)	0
Rash	32 (12)	1 (< 1)	12 (5)	1 (< 1)

[a]Treatment-emergent AE is defined as an adverse event with a start date on or after the date of first dose of study treatment and up to 30 days after date of last dose of study treatment. Patients are counted once if experiencing multiple adverse events. AEs were classified according to the MedDRA systems of preferred terms (version 22.1). [b]Combined neutropenia and neutrophil count decreased. [c] Combined anemia, hemoglobin decreased, red blood cell count decreased. [d]Combined leukopenia, white blood cell count decreased. [e]Combined lymphopenia, lymphocyte count decreased. [f]Combined thrombocytopenia, platelet count decreased.

Abbreviations: AE, adverse event; SG, sacituzumab govitecan; TPC, treatment of physician's choice.

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